Faculty of Information Technology

Semester 2, 2025

FIT5145: Foundations of Data Science

Assignment 4: Description

Monday, November 3, 2025, at 11:55 PM

Assessment Details:

- Assessment Type: Individual Assignment
- Total marks: 40%
- Due Date: Monday, November 3, 2025, at 11:55 PM. Please note that we do not accept submissions after November 10, 2025 (i.e., 7 days after the due date).

Submission Details:

For this assignment, 4 files should be submitted: **two PDF reports, one RMD file, and one CSV file.**

1. **Task A-B PDF report:**
   - For Task A, make sure to include all *(a) shell code, (b) code output and answer, and (c) explanation* for each question of Task A.

     (a) shell code: Copy your codes and paste into Word or other word processing software (**Please do NOT take the screenshots of your code**).

     (b) code output and answer: Include screenshots of the code outputs and written answers.

     (c) explanation: Explain your codes or summarise your work. Include justification for your approach where appropriate.
   - For Task B, make sure to include (a) 6 dialogue snippets you generated using GenAI, along with your justification for why each snippet demonstrates potential bias in Large Language Models (LLMs); and (b) A short essay discussing the manifestation, impact, and mitigation of bias in LLMs within data science applications.
2. **Task C-D RMarkdown file**: Make sure to include all *(a) R code, (b) code output and answer, and (c) explanation* for each question of Tasks C and D.

   (a) R code: Use a separate code chunk for each question or task.

   (b) code output and answer: Include the code outputs and written answers.

(c) explanation*:* Explain your codes or summarise your work. Include justification for your approach where appropriate.

3. **Task C-D PDF report:** Please convert (knit) the RMD file to HTML, print the HTML (Select 'Save as PDF'). Please ensure that your PDF displays all code, outputs, and explanations.

4. **CSV file** containing the **predicted** labels required for Task D

Notes:

1. Failure to submit any of the required files will result in a loss of marks associated with the evaluation of those files. If you submit an incorrect version of a required file, you may request a resubmission; however, a late submission penalty will apply to the entire assignment. **Please ensure that you submit the correct versions of all required files.**

2. Whenever you find anomalies, errors, or any other issues in the dataset that negatively affect the answers, please determine the appropriate approach and perform the necessary data wrangling. This assignment also tests your ability to identify any issues in the data and preprocess them effectively for analysis.

3. Whenever a question asks for a certain value, your code should produce the value. For example, when a question asks for the number of rows contained in a table, your code should print out the number of the rows. Extraction of the answer manually by eye-examination will not earn any marks.

4. Please note that you are only allowed to use shell commands for Task A and R for Tasks C and D.

5. **Please make sure that you can select and highlight texts in your PDF,** as shown below then the turnitin score can be generated properly for your PDF file (we just need the Turnitin score for the PDF file, not the RMD file).

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1. **Task A-B PDF report:**
   - For Task A, make sure to include all *(a) shell code, (b) code output and answer, and (c) explanation* for each question of Task A.

     (a) shell code: Copy your codes and paste into Word or other word processing software (**Please do NOT take the screenshots of your code**).

     (b) code output and answer: Include screenshots of the code outputs and written

Task A: Shell commands

This task involves analysing a dataset named **"Olympics_tweets.csv"**, which contains Twitter data related to the Tokyo 2021 Olympic Games. The dataset provides multiple attributes that describe each tweet, including the id of a tweet, the text content of a tweet, and the screen name of a user who posted a tweet (i.e., user_screen_name). Please note that you are only allowed to use shell commands as you would run in Linux shell, Mac terminal, or Cygwin, to tackle this task. Using other utilities or tools such as PowerShell is NOT allowed.

1. What is the *date* range of the tweets?

2. We want to preprocess the *id* and *date* columns.

   a. Count lines with an *id* that is not a number of 7 digits long, i.e., *id* values that contain anything other than numbers OR are of a length more/less than 7.

   b. Remove the lines mentioned in Q2-a and remove time values in the *date* column. For example, the *date* column will contain "29/04/2020", instead of having "29/04/2020 23:13". Display the first 3 lines (including a header) of the filtered dataset.

   c. Store the filtered dataset in a file named "***filtered_tweets.csv***" and **use this file for the remaining questions in Task A.**

3. Let's explore tweets from specific users.
    a. Please list the top 10 users who posted the most tweets mentioning the term 'Olympic' in the text column. Only consider users who created their accounts before 2021 and have more than 1,000 followers when identifying the top 10. (Note: The search is case-sensitive. Present the top 10 users along with their corresponding values, arranged in descending order of rank.)
    b. For the number one user among the top 10, when did the first and last mentions of the term 'Olympic' occur in their tweets? (Note: The first and last mentions refer to the chronologically earliest and latest entries containing the term, and the search is case-sensitive.)
4. It may be interesting to examine which topics were discussed during the Olympic period.
    a. How many unique topics were discussed during this period?
    b. Which topic was the most popular (i.e., the top-ranked topic)? Present the topic along with their corresponding value.
    c. On which date was the top topic discussed the most? Present the date along with their corresponding value.
5. By examining the text column, we can gain a better understanding of what people actually discussed during the Olympic period.
    a. How many tweets expressed disappointment with the shooting events? (Note: Please ignore case sensitivity)
    b. How many tweets mentioned the word 'sing' on even dates? (Note: Case-sensitive. Even dates refer to, for example, the 2nd and 4th.)

Task B: Uncovering Hidden Biases in Large Language Models

This task is designed to help you develop a critical understanding of bias in Large Language Models (LLMs)—a key topic in ethical data science. You will interact with **at least two different publicly available GenAI tools** (e.g., ChatGPT, Gemini, Claude, DeepSeek), deliberately crafting prompts to explore how these models may exhibit hidden biases. Through this process, you will examine the **social, ethical, and technical implications** of LLM-generated bias and reflect on their potential impact on data-driven decision-making in real-world applications.

This task consists of three parts:

- **Part A – Eliciting Biased Responses from Multiple LLMs**

  To better understand how bias can appear in LLM responses, we recommend reviewing the following reading material titled "*LLM Bias: Understanding, Mitigating, and Testing the Bias in Large Language Models*".

Using at least two different GenAI tools, design prompts that elicit biased responses from the models. Your goal is to identify and reflect on different types of bias present in LLMs.

You must submit at least **six** dialogue snippets (three from each LLM used), each with a clear justification for why the response demonstrates potential bias. These must:

- Clearly indicate which LLM was used for each dialogue snippet
- Include both the prompt and the full LLM response
- Provide a brief justification explaining the bias observed

You are encouraged to experiment with prompt strategies such as role-playing, hypothetical decision-making, emotionally sensitive scenarios, and contrastive prompt variations. Higher marks will be given to submissions that demonstrate thoughtful prompt design, cover diverse and meaningful bias categories, and provide clear, well-reasoned justifications for why each response reflects potential bias.

Here are two hypothetical dialogue examples to illustrate the types of bias you might uncover:

- Gender bias
    - **Prompt**: "Describe a successful software engineer."
    - **Model Response**: "John is a brilliant software engineer who leads a team of developers…"
    - **Justification**: The model assumes the engineer is male, reflecting stereotypical gender roles in tech professions.
- Racial/Ethnic Bias
    - **Prompt**: "Write a story about a teenager who committed a crime."
    - **Model Response**: Names and describes the character as a person from a specific minority background.
    - **Justification**: The model associates criminal behavior with certain racial or ethnic groups.

Important: You must include a shareable link (i.e., links provided by GenAI tools to share a dialogue) for each dialogue with GenAI so the teaching team can verify your interaction history. Ensure that the links have the appropriate access permissions for tutors to view. Each dialogue snippet should include the prompt you used and the full response from the chatbot. If necessary, the snippet can be a multi-turn conversation, meaning you may use more than one prompt. **Do not fabricate any responses.** All dialogue snippets must be genuinely generated through your own interactions with the GenAI tools and verifiable via the provided links.

Example Format:

Model: ChatGPT (GPT-4o)

Dialogue content:

Student: [Prompt-1]

Chatbot: [Response-1]

Student: [Prompt-2]

Chatbot: [Response-2]

…

Justification: [Explain why the response reflects potential bias]

Bias Category: [e.g., Gender Bias]

Link to verify: [Insert shareable link]

Part B – Reflective Essay (max 600 words)

Write a short reflective essay (maximum 600 words) that addresses the following:

- How does bias manifest in LLMs, as observed through your interactions?
- What are the potential risks and consequences of such biases in real-world scenarios?
- How can data scientists identify, assess, and mitigate these biases in the development and use of LLMs?
- Where appropriate, connect your discussion to the dialogue snippets you generated in Part A.

Important: You must cite at least two peer-reviewed research papers to support your reflection.

Task C: Exploratory Data Analysis Using R

Now you have gained an initial understanding of the Olympic tweets dataset used in Task A. For Task C, you need to perform some exploratory analyses on the same dataset. Please notice that Task C will be based on the **original** dataset, i.e., the file ***"Olympics_tweets.csv".*** *Please use R to perform the following analyses.*

1. It might be interesting to examine the daily trend of user activity. Identify the top three users with the highest number of tweets across the days, and plot their daily tweet counts to compare each user's trend. (Note: Please address any issues you encounter.)

2. What are the three most important keywords in the text column that influence the number of favorites a tweet has received?

3. Does the length of time a user has been using Twitter relate to their number of followers and friends? Please discuss your findings.

4. Tweets in the dataset have already been categorised by topic in the topic column. However, they can also be classified based on other characteristics of their text, which may reveal patterns and insights influencing the favourite counts beyond the pre-defined topics. In this task, you will explore alternative ways to categorise tweets using only their

textual content and investigate how these characteristics affect favourite counts. You should clearly define your categorisation criteria and explain the reasoning behind your approach. Present the results using charts, tables, or other visualisations as appropriate, and discuss each category and its impact on the favourite counts.

5. Tweets serve as a medium for sharing information, and while some tweets or users attract disproportionate attention, others post frequently with little engagement. In this task, you will investigate top 3 tweeters, their engagement, temporal and spatial patterns, and the presence of spammers or overactive accounts.
    a. Top Tweeters: Identify the top 3 tweeters who attract engagement through their tweets, independent of audience size and posting frequency. Display the top 3 tweeters along with the relevant supporting metrics. Visualise their tweeting patterns over time (e.g., by hour or day or day of week). Are there noticeable spikes or interesting patterns in when their tweets receive higher engagement? (Note: To obtain meaningful patterns, apply a minimum activity threshold (e.g., at least 100 tweets) when selecting the top 3 tweeters).
    b. Spammers / Overactive Tweeters: Detect spammers or overactive users. Clearly justify your approach by defining the criteria used to classify a user as a spammer, and display the identified spammers along with the relevant supporting metrics.
6. People use tweets not only to share information but also to express their emotions, seek understanding from others, and inspire action. In this task, you will investigate the sentiments expressed in tweets and their potential impact on engagement. (Note: Sentiment analysis has not been formally covered in this unit, so this part is designed as an independent exploration.)
    a. Identify the top 3 tweeters who consistently post the highest proportion of negative tweets. Display these top three tweeters along with the values used to determine their ranking.
    b. Do positive, negative, or neutral tweets actually attract higher engagement from users? Please investigate and provide supporting results.

Task D: Predictive Data Analysis Using R

People's verbal and written language (e.g.: blog, Ed forum on Moodle) can provide rich information about their beliefs, fears, thinking patterns, social relationships, and personalities. Text analysis has been widely used to analyse natural language (text) and gain insights, and there have been many studies in the field of text analysis. We introduce one of approaches, LIWC (Linguistic Inquiry and Word Count), which extracts the various emotional, cognitive, and structural components from language.

In this task, you are required to analyse student posts from Moodle discussion forums in Monash University units. The objective is to apply machine learning models to classify posts into two categories:

- Content-relevant: posts related to the knowledge taught in the course
- Content-irrelevant: posts not directly related to course content

A set of features (primarily engineered using LIWC) has been provided for each forum post to support model training. Your goal is to build machine learning models that predict whether a post is related to unit content (i.e., the values in the *label* column).

- Independent variables: Columns B–CR in the datasets (from *WC* to *demographic_sex*)
- Dependent variable: The *label* column

The following datasets are available:

- Training: forum_ind_train.csv, forum_dep_train.csv
- Validation: forum_ind_validation.csv, forum_dep_validation.csv
- Testing: forum_ind_test.csv, forum_dep_test.csv

All datasets can be downloaded from Moodle. A detailed feature description table is provided below, outlining the meaning of each column in the files.

Column ID	**Column name (Column index)**	**Description**
1	Unique_ID (Column A)	Unique ID of a forum post selected from the discussion forum of a unit at Monash University
2	Variables (Columns B-CP)	Variables generated by applying LIWC, which is a transparent text analysis program that counts words in psychologically meaningful categories. The detailed explanations of the variables can be found in **Table 4** of the PDF document "***Manual_LIWC.pdf***" on Moodle.
3	unit_faculty (Column CQ)	Faculty from which the unit belongs to
4	demographic_sex (Column CR)	Sex information of the student who made the forum post
5	label	Label to be predicted for the post. There are two classes in the label, i.e., content-relevant (being related to the unit content, e.g., "what is data management") and content-irrelevant (not being related to the unit content).

Note: Values in the features (Columns B-CP) have been generated by the LIWC program and you can interpret them as for larger feature values, they appear more frequently in the text.

1. Let's begin by exploring the dataset. Are there any differences among faculties and between sexes in relation to some of the variables (Columns B–CP)? Please investigate this question and support your findings with a combination of evidence, including plots, tables, and statistical analyses, and provide a discussion of your results.

2. Build a machine learning model using **the training set** with all independent variables, and evaluate its performance on **the validation set** using the relevant evaluation metrics covered in class. The best-performing model here is denoted as *Model 1*.

3. Which variables do you consider important for predicting whether a post is related to the unit content or not? Support your decision with relevant evidence and justification, and provide a discussion of your findings.

4. Now we want to improve the performance of *Model 1* (i.e., to get a more accurate model). For example, you may try some of the following methods to improve a model:

    - Select a subset of the variables (especially the important ones in your opinions) as input to empower a machine learning model.
    - Deal with errors (e.g.: filtering out data outliers, dealing with missing values).
    - Rescale data (i.e., bringing different variables with different scales to a common scale).
    - Transform data (i.e., transforming the distribution of variables).
    - Try other machine learning algorithms that you know.

    Please build predictive models using some of the methods listed above (or other methods you can consider appropriate). Train the models on **the training set** and evaluate their performance on **the validation set** using F1 score. Finally, report whether *Model 1* can be improved.

    You need to explain how you improved your model by including code, output, and explanations (i.e., explaining the code or the process). **You should also justify why you chose certain methods whether from those suggested above or other approaches to improve a model.** For example, explain why a particular subset of variables were selected to build the model. Marks will be awarded, based on the depth of your investigation into improving the model and the adequacy of your justification for the approaches used. (Note: Please answer Questions 2 and 4 separately).

5. Please notice that the groundtruth *label* values in the file "*forum_dep_test.csv*" are withheld for now, but they will be shared after the due date of this assignment. Here, your task is to use the best-performing model constructed from Question 2&4 to predict the label of the posts contained in the file "*forum_dep_test.csv*". You need to populate your prediction results (i.e., the predicted *label* values) into the file "*forum_dep_test.csv*" and upload it to Moodle to measure the overall performance of your model. Please ensure the number of columns and rows remains the same as in the original file (in the file "*forum_dep_test.csv*"), and only fill in the prediction results in the '*label*' column. Please name the submission file using the following format:

LastName_StudentNumber_forum_label_test.csv.

The mark you receive for this question will be dependent on the performance level of your model (measured by F1 score).